

Exercise 1

Write a Python program using the `networkx` to create and visualize an undirected graph with the following specifications:

1. Create a graph with 5 nodes labeled `A`, `B`, `C`, `D`, and `E`.

2. Add the following edges between nodes:

* "A" — "C"

* "B" — "C"

* "B" — "D"

* "C" — "D"

* "C" — "E"

* "D" — "E"

3. Set the following 2D coordinates for each node:

* A: (1, 5)

* B: (4.5, 6.5)

* C: (3.5, 1.4)

* D: (5.8, 3.5)

* E: (7.8, 3.8)

4. Visualize the graph with these style options:

* Display node labels

* Bold white font color for labels

* Red-colored nodes with size 1000

* Font size of 15

* Edge width of 4

Solution:

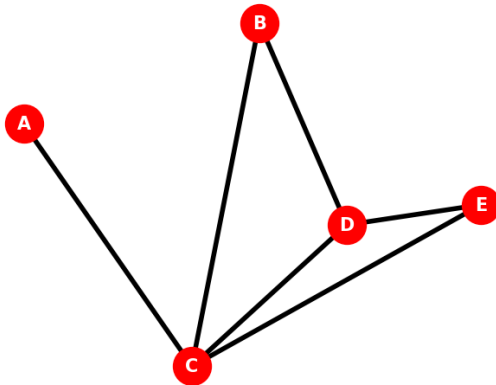
```
import networkx as nx
import numpy as np
```

```
G=nx.Graph()
G.add_node("A")
G.add_node("B")
G.add_node("C")
G.add_node("D")
G.add_node("E")
```

```
G.add_edge("A","C")
G.add_edge("B","C")
G.add_edge("B","D")
G.add_edge("C","D")
G.add_edge("C","E")
G.add_edge("D","E")
```

```
pos={
    "A":(1,5),
    "B":(4.5,6.5),
    "C":(3.5,1.4),
    "D":(5.8,3.5),
    "E":(7.8,3.8)
}
```

```
nx.draw(G,with_labels=True,font_weight='bold',font_color='white',node_color='red',node_size=1000,font_size=15,width=4,pos=pos)
```



Exercise 2

Write a Python program using the `networkx` library to create and visualize a directed graph with the following requirements:

1. Create a directed graph with 5 nodes labeled `P`, `Q`, `R`, `S`, and `T`.

2. Add the following directed edges:

- * "P" → "Q"
- * "P" → "R"
- * "Q" → "S"
- * "R" → "S"
- * "S" → "T"
- * "Q" → "T"

3. Set the following 2D coordinates for each node:

- * `P`: (2, 4)
- * `Q`: (5, 5.5)
- * `R`: (3.5, 2)
- * `S`: (6, 3)
- * `T`: (8, 3.5)

4. Visualize the directed graph with the following visual styles:

- * Display node labels
- * Bold Black font color for labels
- * Yellow-colored nodes with size 800
- * Font size of 13
- * Edge width of 3

```
import networkx as nx
```

```
G = nx.DiGraph()
```

```
nodes = ["P", "Q", "R", "S", "T"]
```

```
G.add_nodes_from(nodes)
```

```
edges = [("P", "Q"), ("P", "R"), ("Q", "S"), ("R", "S"), ("S", "T"), ("Q", "T")]
```

```
G.add_edges_from(edges)
```

```
pos = {  
    "P": (2, 4),  
    "Q": (5, 5.5),  
    "R": (3.5, 2),  
    "S": (6, 3),  
    "T": (8, 3.5)  
}
```

```
nx.draw(G,with_labels=True,font_weight='bold',font_color='white',node_color='red',node_size=1000,font_size  
=15,width=4,pos=pos)
```

